Freight rate prediction

Spotter · Machine Learning Engineer assessment

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8 October 2026

The result in one slide

$37.5

MAE on the Sep–Oct holdout (clean rows), MAPE 1.61%

2.35% → 1.61%

holdout MAPE: CatBoost with a time coordinate → level decomposition

$832–884

December chart: weekly wave, gentle trend, quarter-end ramp

- **Validation:** forward in time only; every choice made on two early folds; the Sep–Oct fold scored once.
- **Model:** log rate per mile = market level $L(\text{date})$ + CatBoost for the cross-section.
- **Tuning** changed little: 51 Optuna trials, CatBoost defaults kept.

Problem and data

The task

Predict `posted_rate`, the USD price of one truck load, from:

- **Lane:** pickup and delivery city, their coordinates, distance
- **Load:** equipment (Dry Van / Reefer / Flatbed), weight
- **Time and market:** date, `market_index` (daily, market-wide), `quote_signal` (weak, per load)

Data

- **Labelled:** 48,000 loads, Jan–Oct 2025
- **Predict:** 12,000 loads, Nov–Dec 2025; the metric is undisclosed
- **December chart:** Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb; one prediction per day

Key EDA findings

- **Out of time:** validation is Nov–Dec and labels end on 31 Oct $\Rightarrow$ time-based splits only.
- **Unseen cities:** 8 cities appear only in validation (12.1% of its rows) $\Rightarrow$ use coordinates and distance, not city IDs alone.
- `market_index` is one market-wide value per day with a weekly cycle (Thu peak, Sun trough); correlation 0.64 with the daily median rate per mile.
- `quote_signal` is weak (correlation ≈ 0.1), with no sign of leakage.
- **Drift:** the rate level rises $\approx 6\%$ from Jan to Oct at a similar `market_index`; quarter-end spikes.
- **Low noise floor:** within lane $\times$ equipment $\times$ week, rate per mile varies by $\approx 2.2\%$ (CV).

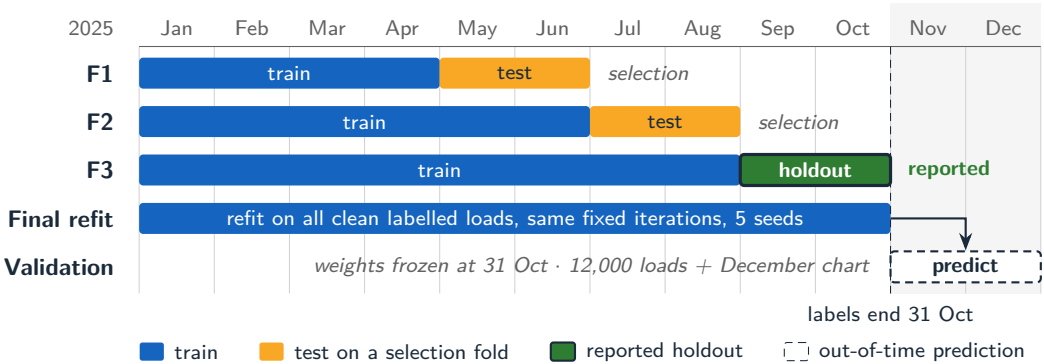
Data quality: what we fix and how

Issue (48,000 labelled loads)	Rows	Handling
Corrupted rate: $\times$2.5–5.5 or $\times$0.2–0.45	677 (1.4%)	drop (training, clean metrics)
Negative weight (a sign flip)	292 (0.6%)	absolute value + flag
Weight at the 47,500 lb cap / 5,000 lb floor	1,204 / 32	keep the value + flag
Missing weight	300 (0.6%)	keep NaN + flag
Missing `market_index`	374 (0.8%)	that day's mean + flag
Distorted coordinates, consistent per city	all	use as internal geometry

Corrupted labels: $|\log(\text{rpm/lane reference})| > \log 1.6$, in an empty gap (genuine < 0.23, corrupted > 0.73).

Validation

Validation design: forward in time only



Validation is out of time: every fold trains on the past and tests on the next two months.

Split protocol

- **Select** on F1 + F2 only: the model design, the features and the hyperparameters.
- **Report** F3 (Sep–Oct) once, with the final choice.
- **Refit** on all clean Jan–Oct loads (47,323), 5 seeds; predict Nov–Dec.
- **Fixed iterations:** no early stopping on a reported fold.
- **Seeds:** 2–3 per configuration, mean ± sd.

Cold-start slice

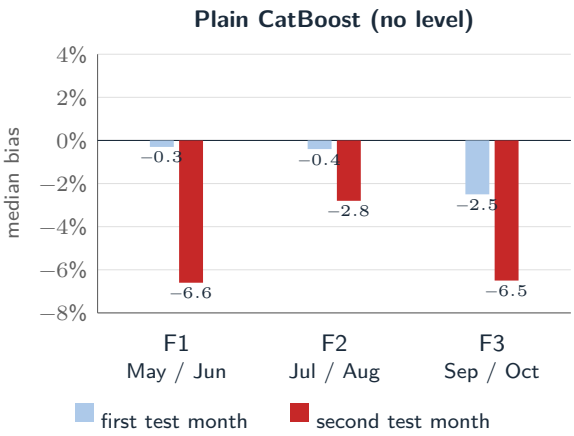
On F3, remove every training row that touches 8 random cities (2 draws): held-out cities \$40.3 MAE vs \$34.4 for the rest.

Metrics

MAE in \$ on clean rows (primary), plus MAPE, all-row MAE and RMSE; month bias as a diagnostic.

Model and results

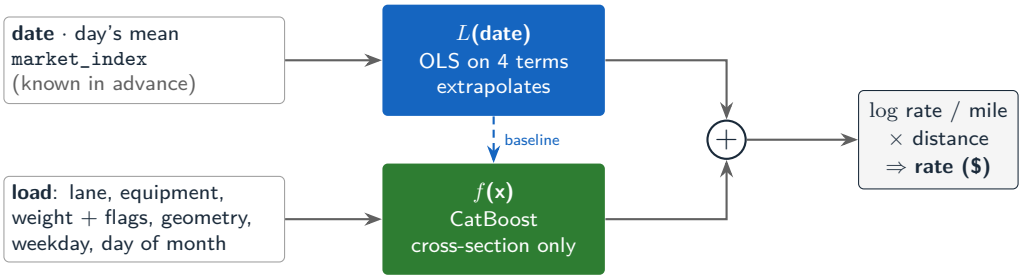
Plain trees miss the second month



- **First test month:** $\approx$ 1% MAPE (May, July).
- **Second month:** 5.7–6.5% MAPE, up to 6.6% too low.
- Trees **cannot extrapolate** the rising price level.
- **Month proxies** (quote signal's monthly level, days to quarter end, trailing averages) let them reuse old levels.
- Adding time t: F3 MAPE 2.35%, but June still -7.2%.

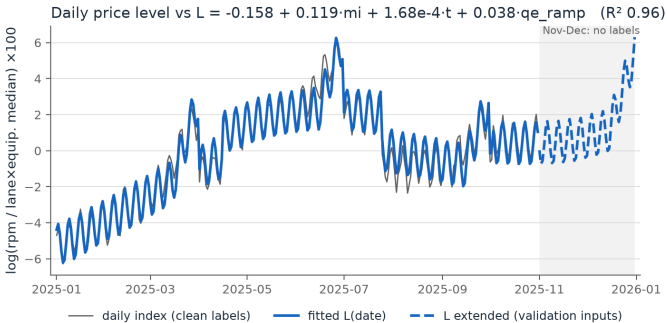
The fix: split the price into level and cross-section

$$\log \frac{\text{rate}}{\text{distance}} = \underbrace{\beta_0 + \beta_{\text{mi}}\,\bar{m}(d) + \beta_t\, t + \beta_{\text{qe}}\, q(d)}_{L(\text{date})\text{: market level, OLS}} + \underbrace{f(\mathbf{x})}_{\text{CatBoost}}$$



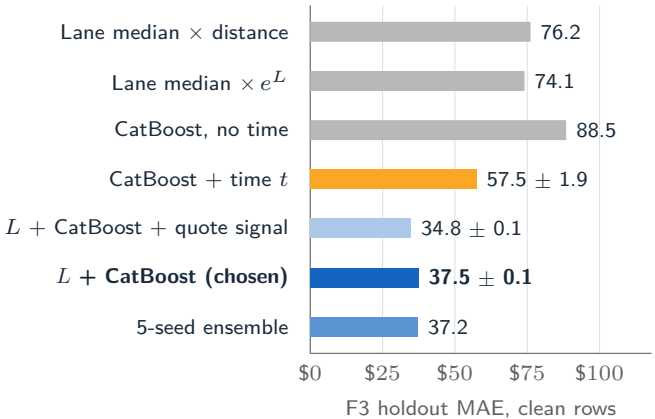
$\bar{m}$: day's mean market index · t: days since 1 Jan · q: quarter-end ramp · quote signal dropped from the trees

The market level $L(\text{date})$



- **Weighted R^2 0.96** over 304 days.
- **Market index:** $+0.1 \Rightarrow +1.2\%$.
- **Trend:** $+0.51\%$ per 30 days.
- **Quarter-end ramp:** $+3.85\%$ on a quarter's last day.
- Every term is **known in advance**, so L extrapolates.

Results on the F3 holdout (Sep–Oct)



- **$37.5 MAE, 1.61% MAPE:** half the yardstick's error.
- **Seed noise:** $0.1–0.5.
- **Month bias:** -0.7% / -1.1%, was -2.5% / -6.5%.
- **Error ≈ 1.6%** of the rate at every distance; Flatbed hardest.

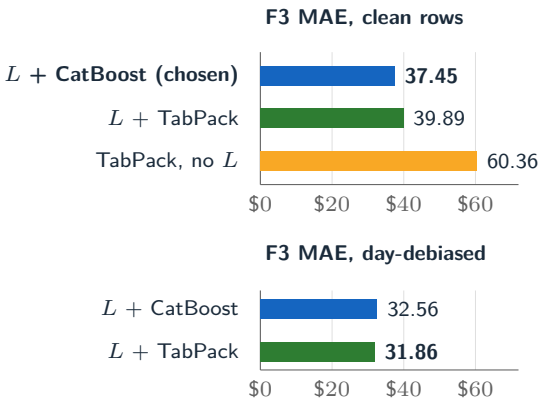
Tuning: the hyperparameters barely matter

F1+F2, 3 seeds	day-debiased	plain	fit (s)
best trial (#36)	32.57	50.53	65.5
CatBoost defaults	32.91	50.63	23.7
gain	0.34	0.10	$\times 2.8$

MAE in \$. Day-debiased: each test day's median error removed, so the score isolates the trees.

- **Ablation:** masking weight costs \$26.6, coordinates \$1.9, every other family $<$ \$0.5.
- **Optuna:** 51 trials, TPE + median pruner.
- Gain inside the \$0.5 tie rule, fits $\approx 3\times$ slower $\Rightarrow$ **defaults kept**.
- The decomposition moved MAPE $2.35\% \rightarrow 1.61\%$.

TabPack: a stronger learner is not the bottleneck

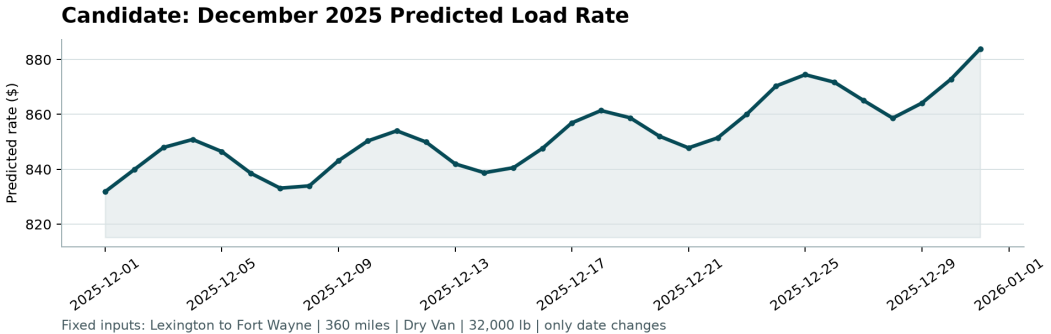


Mean over 3 seeds. Day-debiased: each test day's median error removed (cross-section only).

- **TabPack** (ICML 2026): 64 MLPs, sampled hyperparameters, greedy online ensemble; paper config, 38 min on a rented GPU.
- **Without L:** second month -3.0%, the plain-trees failure.
- **In CatBoost's slot:** cross-section \$0.7 better, level offset worse.
- **Blend on F1+F2:** $w = 1$, CatBoost alone; gate fails $\Rightarrow$ **submission unchanged**.

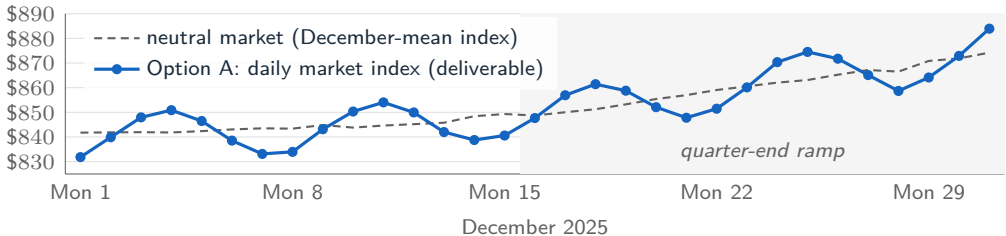
December chart

December 2025: Lexington → Fort Wayne



Fixed lane: 360 mi, Dry Van, 32,000 lb; only the date changes. $831.84 on Mon 1 Dec to $883.92 on Wed 31 Dec, mean $852.88.

What shapes December



- **Inputs, not labels:** each day's mean `market_index` from `validation.csv`.
- **Weights frozen** at 31 Oct.
- **Wave** $\approx$ \$19; **trend** $+0.51\%$ / 30 days; **ramp** $+3.85\%$ by 31 Dec.
- **Neutral market:** within \$11.4, same mean.

Summary

Takeaways and next steps

Takeaways

- Validate the way the model will be used: forward in time, F3 scored once.
- **Structure beat tuning:** the level decomposition took F3 MAPE from 2.35% to 1.61%.
- **The learner is not the bottleneck:** TabPack was worth $\approx$ \$1, the decomposition $\approx$ \$20.
- Every data fix is explicit and flagged.
- December: weekly wave + trend + quarter-end ramp.

Limits $\rightarrow$ next steps

- Holidays not modelled $\rightarrow$ holiday priors.
- Trend risk bounded: freezing t lowers mid-Dec by $\approx 0.8\%$.
- β_{mi} and β_t trade off over short windows.
- Flatbed weakest; metric unknown.
- Refit TabPack on train + val, or blend it on a level-corrected target; monotone weight constraint.

Backup: code map

File	Key names	Role
`src/freight/outliers.py`	`RpmReference`, `flag_corrupted`	robust lane reference, corrupted labels
`src/freight/split.py`	`Fold`, `get_folds`, `cold_start_holdout`	forward-time folds, cold-start cities
`src/freight/level.py`	`MarketLevel`, `qe_ramp`	the market level $L(\text{date})$
`src/freight/models.py`	`LevelCatBoostRate`	CatBoost with L as baseline
`src/freight/final.py`	`fit_final`, `predict_december`	final fit, deliverables
`scripts/tabpack/`	`run_all.sh`, `blend.py`	TabPack check: runs, tables, blend + gate
`Makefile`	`make all`, `make research`	tests $\rightarrow$ train $\rightarrow$ predict $\rightarrow$ score $\rightarrow$ sanity